IsoperimetricInequality: A Lean 4 formalization of the planar isoperimetric inequality via Fourier series

6 September 2026

Summary

`IsoperimetricInequality` is a formalization, in the Lean 4 proof assistant (Moura and Ullrich 2021) and its mathematical library Mathlib (The Mathlib Community 2020), of the classical planar isoperimetric inequality: among all simple closed C^1 curves of a given perimeter L, none encloses more area than the circle of that perimeter, so the enclosed area A satisfies

$$A \leq \frac{L^2}{4\pi}.$$

The development follows the Fourier-analytic argument of Adolf Hurwitz (Hurwitz 1902; Osserman 1978). It reparametrizes a curve over $[0, 2\pi]$, expands its coordinate functions as trigonometric (Fourier) series, and applies Parseval's identity together with Wirtinger's inequality and the arithmetic mean–geometric mean inequality to bound the area. Every step is checked by Lean's kernel against Mathlib, so the result is a machine-verified theorem with no appeal to informal reasoning.

The repository is organized into two files. `IsoperimetricInequality/Basic.lean` builds the required real Fourier analysis from first principles: it defines the trigonometric series $f(x) = \frac{a_0}{2} + \sum_{n\geq 1}(a_n \cos nx + b_n \sin nx)$, proves the orthogonality relations for $\{\cos nx, \sin nx\}$ on $[-\pi, \pi]$, establishes uniform convergence and term-by-term differentiability of the series under an absolutely summable coefficient hypothesis (a Weierstrass M-test argument), and derives **Parseval's theorem** (`Parsevals_thm`) and the integral form of **Wirtinger's inequality** (`Wirtingers_inequality`). `IsoperimetricInequality/Adolf_Hurwitz_proof.lean` defines a structure `SimpleClosedC1Curve` for simple closed C^1 curves, the shoelace `area` functional, `arcLength`/`perimeter`, and the predicate `IsArcLengthParametrized`, and

then assembles the chain of lemmas (`area_parametrized` → `area_simplified` via integration by parts → `area_inequality` via AM–GM → `addition_ineq` via Wirtinger → `isoperimetric_inequality`) that yields the final bound `area  0 .length  .length ^ 2 / (4 * Real.pi)`.

Statement of need

The isoperimetric inequality is one of the oldest problems in mathematics and a standard example in courses on the calculus of variations, Fourier analysis, and differential geometry (Osserman 1978). Despite this prominence, and despite several distinct classical proofs, the planar case had not been available as a reusable, fully checked theorem in Mathlib. Formalized mathematics libraries grow by accumulating exactly such landmark results together with the intermediate infrastructure they require, and this project contributes both.

The reusable infrastructure is the main point of need. Mathlib's existing Fourier material is centred on the complex exponential basis e^{inx} and the L^2 theory of `fourierBasis`. The present development instead works with the real $\{\cos, \sin\}$ series as a concrete pointwise-convergent object, proves that the series may be differentiated term by term, and gives Parseval and Wirtinger identities in the elementary $\int_{-\pi}^{\pi}$ form that appears in textbooks (Stein and Shakarchi 2003). This formulation is directly usable in downstream arguments about periodic functions, boundary-value problems, and inequalities of Poincaré type, independently of the isoperimetric application.

More broadly, the project serves as a worked, self-contained case study for students and practitioners of interactive theorem proving: it shows how a multi-step analytic proof, combining measure theory, uniform limits of derivatives, infinite sums (`tsum`), and elementary inequalities, is structured and discharged in Lean 4, and it can be read alongside the informal proof as a teaching resource. The formalization is continuously integrated (the Lean build, Mathlib cache, and API documentation are produced by GitHub Actions) and is archived on Zenodo (Samarakkody 2026).

Functionality and scope

The top-level theorem `isoperimetric_inequality` is stated for a `SimpleClosedC1Curve 2` and concludes $A \leq L^2/(4\pi)$ for the shoelace area A. Its proof reduces the geometric statement to the analytic core through the lemma chain listed above, and the Fourier facts it consumes — the Parseval and Wirtinger integral identities for the reparametrized coordinate function, its zero-mean (centroid) normalization, and the arc-length constraint $(f')^2 + (g')^2 = (L/2\pi)^2$ — are provided as explicit hypotheses that are instances of the general theorems proved in `Basic.lean`. This keeps the geometric and analytic layers cleanly separated.

Current limitations, which also mark natural directions for future work:

- The characterization of the equality case (the bound is attained **iff** the curve is a circle) is not yet formalized.
- The coordinate Fourier expansion is supplied through the Parseval/Wirtinger hypotheses of the main theorem rather than constructed from a convergence theorem for the specific reparametrized curve; connecting the two would remove those hypotheses.
- The result is restricted to C^1 curves; the rectifiable / bounded-variation generalization is left open.

The development targets Lean `v4.27.0` and Mathlib `v4.27.0`, builds with `lake build`, and its documentation is generated with `doc-gen4`.

Acknowledgements

We thank the Lean community and the Mathlib maintainers for the library and tooling that made this formalization possible.

References

Hurwitz, Adolf. 1902. "Sur Le Problème Des Isopérimètres." *Comptes Rendus de l'Académie Des Sciences de Paris* 132: 401–3.

Moura, Leonardo de, and Sebastian Ullrich. 2021. "The Lean 4 Theorem Prover and Programming Language." In *Automated Deduction – CADE 28*, edited by André Platzer and Geoff Sutcliffe, vol. 12699. Lecture Notes in Computer Science. Springer. https://doi.org/10.1007/978-3-030-79876-5_37.

Osserman, Robert. 1978. "The Isoperimetric Inequality." *Bulletin of the American Mathematical Society* 84 (6): 1182–238. https://doi.org/10.1090/S0002-9904-1978-14553-4.

Samarakkody, Miraj. 2026. *Formalizing the Classical Isoperimetric Inequality in the Two-Dimensional Case*. Zenodo. https://doi.org/10.5281/zenodo.20258500.

Stein, Elias M., and Rami Shakarchi. 2003. *Fourier Analysis: An Introduction*. Vol. 1. Princeton Lectures in Analysis. Princeton University Press.

The Mathlib Community. 2020. "The Lean Mathematical Library." *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–81. https://doi.org/10.1145/3372885.3373824.